National Transportation Safety Board
Aviation Accident Final Report

Location:	Alton, NH	**Accident Number:**	NYC08CA068
Date & Time:	12/22/2007, 1230 EST	**Registration:**	N3070A
Aircraft:	Cessna 170B	**Aircraft Damage:**	Substantial
Defining Event:		**Injuries:**	4 None
Flight Conducted Under:	Part 91: General Aviation - Personal		

Analysis

The pilot stated that he took three friends on a short cross-country pleasure flight. The airplane was entering a downwind leg at the destination airport, when the pilot decided to make a low pass over an adjacent lake, near a friend's property. The pilot said that his depth perception was negatively affected due to the flat light and snow on the ground. While maneuvering over the lake, the right main landing gear contacted the snowy frozen surface, which resulted in the right wing impacting the surface. The airplane came to rest upright on the frozen lake, and sustained substantial damage to the right wing and firewall. The reported weather at an airport about 10 miles north of the accident site, about the time of the accident, included calm winds and an overcast ceiling at 2,200 feet.

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be: The pilot's failure to maintain clearance above a frozen lake while maneuvering. Factors contributing to the accident were the pilot's attempt of the low altitude maneuver, and unfavorable lighting conditions.

Findings

Occurrence #1: IN FLIGHT COLLISION WITH TERRAIN/WATER
Phase of Operation: MANEUVERING

Findings
1. TERRAIN CONDITION - WATER,FROZEN
2. (C) ALTITUDE/CLEARANCE - NOT MAINTAINED - PILOT IN COMMAND
3. (F) LIGHT CONDITION - OTHER
4. (F) LOW ALTITUDE FLIGHT/MANEUVER - ATTEMPTED - PILOT IN COMMAND

Factual Information

Pilot Information

Certificate:	Private	**Age:**	48, Male
Airplane Rating(s):	Single-engine Land; Single-engine Sea	**Seat Occupied:**	Left
Other Aircraft Rating(s):	None	**Restraint Used:**	Seatbelt
Instrument Rating(s):	None	**Second Pilot Present:**	No
Instructor Rating(s):	None	**Toxicology Performed:**	No
Medical Certification:	Class 3 With Waivers/Limitations	**Last FAA Medical Exam:**	05/01/2007
Occupational Pilot:		**Last Flight Review or Equivalent:**	05/01/2007
Flight Time:	1700 hours (Total, all aircraft), 1673 hours (Total, this make and model), 1700 hours (Pilot In Command, all aircraft), 95 hours (Last 90 days, all aircraft), 11 hours (Last 30 days, all aircraft), 0 hours (Last 24 hours, all aircraft)		

Aircraft and Owner/Operator Information

Aircraft Make:	Cessna	**Registration:**	N3070A
Model/Series:	170B	**Aircraft Category:**	Airplane
Year of Manufacture:		**Amateur Built:**	No
Airworthiness Certificate:	Normal	**Serial Number:**	25714
Landing Gear Type:	Tailwheel	**Seats:**	4
Date/Type of Last Inspection:	03/01/2007, Annual	**Certified Max Gross Wt.:**	2200 lbs
Time Since Last Inspection:	98 Hours	**Engines:**	1 Reciprocating
Airframe Total Time:	2973 Hours at time of accident	**Engine Manufacturer:**	Continental
ELT:	Installed, not activated	**Engine Model/Series:**	C145-2A
Registered Owner:	John A. Kiley	**Rated Power:**	145 hp
Operator:	John A. Kiley	**Operating Certificate(s) Held:**	None

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual Conditions	**Condition of Light:**	Day
Observation Facility, Elevation:	LCI, 545 ft msl	**Distance from Accident Site:**	10 Nautical Miles
Observation Time:	1235 EST	**Direction from Accident Site:**	350°
Lowest Cloud Condition:		**Visibility**	10 Miles
Lowest Ceiling:	Overcast / 2200 ft agl	**Visibility (RVR):**	
Wind Speed/Gusts:	Calm /	**Turbulence Type Forecast/Actual:**	/
Wind Direction:		**Turbulence Severity Forecast/Actual:**	/
Altimeter Setting:	30.59 inches Hg	**Temperature/Dew Point:**	-5°C / -6°C
Precipitation and Obscuration:			
Departure Point:	Sanford, ME (SFM)	**Type of Flight Plan Filed:**	None
Destination:	Alton, NH (NH15)	**Type of Clearance:**	None
Departure Time:	1200 EST	**Type of Airspace:**	

Airport Information

Airport:	Locke Lake Airport (NH15)	**Runway Surface Type:**	
Airport Elevation:		**Runway Surface Condition:**	
Runway Used:	NA	**IFR Approach:**	None
Runway Length/Width:		**VFR Approach/Landing:**	None

Wreckage and Impact Information

Crew Injuries:	1 None	**Aircraft Damage:**	Substantial
Passenger Injuries:	3 None	**Aircraft Fire:**	None
Ground Injuries:	N/A	**Aircraft Explosion:**	
Total Injuries:	4 None	**Latitude, Longitude:**	43.386667, -71.233333

Administrative Information

Investigator In Charge (IIC):	Robert J Gretz	**Report Date:**	02/28/2008
Additional Participating Persons:	Mark Auclair; FAA/FSDO; Portland, ME		
Publish Date:			
Note:	This accident report documents the factual circumstances of this accident as described to the NTSB.		
Investigation Docket:	NTSB accident and incident dockets serve as permanent archival information for the NTSB's investigations. Dockets released prior to June 1, 2009 are publicly available from the NTSB's Record Management Division at pubinq@ntsb.gov, or at 800-877-6799. Dockets released after this date are available at http://dms.ntsb.gov/pubdms/.		

The National Transportation Safety Board (NTSB), established in 1967, is an independent federal agency mandated by Congress through the Independent Safety Board Act of 1974 to investigate transportation accidents, determine the probable causes of the accidents, issue safety recommendations, study transportation safety issues, and evaluate the safety effectiveness of government agencies involved in transportation. The NTSB makes public its actions and decisions through accident reports, safety studies, special investigation reports, safety recommendations, and statistical reviews.

The Independent Safety Board Act, as codified at 49 U.S.C. Section 1154(b), precludes the admission into evidence or use of any part of an NTSB report related to an incident or accident in a civil action for damages resulting from a matter mentioned in the report. A factual report that may be admissible under 49 U.S.C. § 1154(b) is available here.